cutable malicious code that latches on to existing, or new, wireless applications. Application-based threats are potentially present anytime a software program is downloaded to, or executed on, a wireless device particularly when the program is downloaded or received from an unknown source. In the wired world, these threats are roughly analogous to the early viruses borne by executable programs (which were later superseded by the rise in Macro viruses—malicious code borne by non-executable files).

The first malicious application-based program that specifically targeted the Palm operating system (OS) used in Palm Pilot personal digital assistants (PDAs) was called ‘Liberty Crack’. The free software, which could be downloaded from a Web site or accessed via Internet relay chat (IRC) rooms, pretended to convert the shareware Liberty Game Boy program into a registered version. When the program was executed, the user was not aware that, in the background, the program was deleting all executable applications in the handheld device. Liberty Crack did not affect the underlying Palm operating system or the embedded applications.

Liberty Crack and similar ‘Trojan horses’ are likely to spread very slowly ‘in the wild’ (i.e., in the real world) and represent a relatively low threat. Liberty Crack is designated a Trojan horse as it masquerades with one purpose, while harbouring a surprise purpose (similar to the Trojan horse of ancient Greece in which soldiers hid inside a hollow wooden horse presented as a gift by the Trojans).

While actual incidences of Liberty Crack have not been encountered in the wild, this Trojan horse is significant in its proof of concept—demonstrating that malicious code can be downloaded and may adversely impact PDAs. Many analysts have labelled Liberty Crack, which first made news in late August 2000, as a harbinger of more malicious code to come. For example, future wireless Trojans could steal data such as address book information, portal passwords, and other confidential information.

An independent developer for Palm computers, know as “Ardiri,” assumed credit for designing Liberty Crack, saying its orig-